

HDI Predictor Documentation

Solution Architecture

Date	16 June 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
Prepared By	Kankipati Sai Venkat

Solution Architecture Diagram



Solution Architecture

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web form where users input the four HDI indicators and view the returned tier/score	HTML5, CSS3, Jinja2 templates
Application Layer	Flask routes that receive form submissions, orchestrate preprocessing and inference, and return rendered results	Python, Flask
Preprocessing	Cleans and scales incoming feature values to match the format used during model training	Pandas, scikit-learn StandardScaler/preprocessing utilities
ML Inference	Loads the serialized RandomForestClassifier and produces the predicted class and probabilities	scikit-learn, joblib/pickle
Data Layer	Stores the training dataset and the serialized model artifact	CSV file, model.pkl
Deployment	Packages the Flask app, model, and dependencies into a single reproducible unit	Docker, Dockerfile, requirements.txt